



Department of Computer Science and Engineering

Academic Year 2023– 2024 (Odd/Even Semester)

Degree, Semester & Branch: B.Tech, III & CSBS

Course Code & Title: AD3491 & Fundamentals of Data Science and Analytics

Name of the Faculty member(s): Dr. M. Gomathy Nayagam

Innovative Practice Description

- **Unit/ Topic: Unit II/ Linear Least Square**
- **Course Outcome: CO5**
- **Topic Learning Outcome: TLO10**
- **Activity Chosen: Problem based Learning**
- **Justification:**

The topic “Linear Least Squares” is one of the mathematical model to develop the Machine Learning model with Linear regression. To understand the concept of mathematical modelling of linear regression, this problem-based learning activity will provide a meticulous understanding of the concept.

- **Time Allotted for the Activity: 20 Minutes**
- **Details of the Implementation:**
 - Course instructor explained the basic concept and he explained mathematical model of Linear Least square Regression with an example to the students for 20 Minutes in the class
 - After completion of the class, the course instructor gave the following problems to the students to solve it using linear regression model. The question for the problem is:

“Sam found how many hours of sunshine vs how many ice creams were sold at the shop from Monday to Friday. The data for the hours of sunshine and how many ice creams were sold at the shop from Monday to Friday is given in the following table

“x” Hours of Sunshine	“y” Ice cream sold
2	4
3	5
5	7
7	10
9	15

Find the line of best fit for the above data using Linear regression: Least Square”

- Based on the knowledge gathered during the class, the students solved the problem and shown the answer to the course instructor separately. Students individually disclosed the result at each step. The course instructor verified that answer and he made correction to the students those who done the problem incorrectly.

• **CO–PO/ PSOMapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 5	3	3	3	2	2	1	-	1	1	1	1	1	3	3	1

(1– Low 2–Moderate 3 – High)

• **PO/ PSOMapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5
	2	1	2	1	1
Justification for correlation	The students could be able to apply the basic mathematical knowledge to solve the given problem	The students could be able to formulate the solution for given problem using Least square method- Linear regression	The students could be able to provide the solution for the given problem using Linear regression	The students could be able to build the Linear regression ML model for the problem	The students could be able to use the appropriate ML packages to solve the problem

Innovative practice	PO9	PO10	PO11	PO12	PSO1	PSO2
	2	1	1	3	1	1
Justification for correlation	The students could be able to solve the problem using Linear regression individually	The students could be able to communicate their results of the problem and discuss	The students could be able to apply the knowledge of Linear regression in project.	The students could be able to learn the newer package to build the ML model	The students could be able to build the Linear regression model for Mobile application problems.	The students could be able to develop the ML model for any IT problems and provide the solution.

		with their peer team members				
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Innovative practice	PSO1	PSO2	PSO3
	3	3	1
Justification for correlation	The students could be able to apply the Linear Least Machine learning technology	The students could able to develop data science application	The students could be able to solve the any real-world problem using Linear regression individually

- **Images/Screen shot of the practice:**

The sample photo copies of students' notebook (Ms.P.Keerthana, Ms.Harini and Ms. Haritha) are attached.

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students are very much enjoyed and actively solved the problem with the discussion of both their peer students and course instructor.

- ❖ **Benefit of the Practice (E.g. Outcome Attainment would have increased due to innovative practice over conventional practice)**

- It helps students to solve the real time problem using linear regression mathematically.
- It helps the students to understand the importance of mathematical concept for building ML model.
- It motivates the students to improve their critical thinking, problem solving and oral communication skills.

- ❖ **Challenges faced in implementation:**

- Make all students to solve the problems
- Make the students to answer the result of each student was difficult
- Individual assessment/observation was difficult.

References:

- ❖ [https://citl.illinois.edu/citl-101/teaching-learning/resources/teaching-strategies/problem-based-learning-\(pbl\)](https://citl.illinois.edu/citl-101/teaching-learning/resources/teaching-strategies/problem-based-learning-(pbl))
- ❖ <https://www.hunschool.org/resources/problem-based-learning>

- ❖ Mr.K.VigneshSaravanan, “Standard Error Estimation” in course CS3352Foundation of Data Science,https://www.ritrjpm.ac.in/images/computer-science/2022-2023/3.KVS_Problem_based_Learning_FDS.pdf
- ❖ Ms.G.SakthiPriya, “Case Study-the Next Gen POS”, in course CS8592 Object Oriented Analysis and Design, https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1_GS_CS8592_problem_based_learning.pdf

Signature of Faculty Member
(Dr.M.GomathyNayagam)

HOD

P. Karthana
CSBS

x	y
7	50
3	19
20	80
9	55
11	66
14	72
1	4
4	36
12	70
	35

$$\bar{x} = 8.4$$

$$\bar{y} = 48.7$$

Data x:
7, 3, 20, 9, 11, 14, 1, 4, 12, 3

Data x in ascending order: 1, 3, 3, 4, 7, 9, 11, 12, 14, 20

Rank x: 1, 2.5, 2.5, 4, 5, 6, 7, 8, 9, 10

Data y : 50, 19, 80, 55, 66, 72, 4, 36, 70, 35

Data y in ascending order :

Rank y : 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Data x	Data y	Rank x	Rank y	d	d ²
7	50	5	5	0	0
3	19	2.5	2	0.5	0.25
20	80	10	10	0	0
9	55	6	6	0	0
11	66	7	7	0	0
14	72	9	9	0	0
1	4	1	1	0	0
4	36	4	4	0	0
12	70	8	8	0	0
3	35	2.5	3	0.5	0.25

$$P = 1 - \frac{6 \sum d^2}{n(n^2-1)} = 1 - \frac{6 \times 0.5}{10(10^2-1)} = 1 - \frac{3}{990} = 1 - 0.00303 = 0.997$$

Very Strong positive monotonic correlation between data y & data x

Linear regression model: Least square

• To find the best fit for N points:

• Step 1: for each (x, y) point calculate x^2 and xy

• Step 2: Sum all x, y, x^2 and xy , which gives as $\Sigma x, \Sigma y, \Sigma x^2$ and Σxy

($\Sigma \rightarrow$ means sum up)

• Step 3: calculate Slope m

$$m = \frac{N \Sigma(xy) - \Sigma x \Sigma y}{N \Sigma(x^2) - (\Sigma x)^2}$$

($N \rightarrow$ No. of points)

Step 4: calculate intercept

$$b = \frac{\sum y - \frac{\sum x \sum y}{N}}{N}$$

Step 5: Assemble the equation of line

Sam found how many hours of sunshine vs how many ice creams were sold at the shop from Monday to Friday

X Hours of Sunshine	Y Ice Cream Sold
2	4
3	5
5	7
7	10
9	15

Step 1

calculate x^2 and xy

x^2	xy
4	8
9	15
25	35
49	70
81	135

Step 2: sum of all $\sum x$, $\sum y$, $\sum x^2$, $\sum xy$

$$\sum x = 26, \sum y = 41, \sum x^2 = 168, \sum xy = 263$$

Step 3

$$m = \frac{N \sum (x, y) - \sum x \sum y}{N \sum (x^2) - (\sum x)^2} = \frac{5 \times 263 - 26 \times 41}{5 \times 168 - 26^2} = 1.5183$$

Standard deviation: (Square root of variance)

→ Notations: SD, s

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

Eg:-

7, 5, 9, 13, 3, 11, 15, 9. To find the SD of the number of holiday consumers.

1. Find the mean, $\bar{x}$, of your data

$$(7 + 5 + 9 + 13 + 3 + 11 + 15 + 9) / 8 \Rightarrow 9$$

2. Find the deviation from the mean

$$\begin{matrix} 7-9 & 9-9 & 3-9 \\ -2, & 0, & -6, & 4, & -6, & 2, & 6, & 0 \end{matrix}$$

3. Note, however, that the sum of deviations is zero.

population S.D

$$\sigma = \sqrt{\sigma^2}$$

sample S.D

$$s = \sqrt{s^2}$$

Eg:-

Sample data = 3, 5, 7 S.D = ?

Sample mean = 5

$$S = \sqrt{\frac{(3-5)^2 + (5-5)^2 + (7-5)^2}{3-1}} = \sqrt{4} \Rightarrow 2$$

3-1

✓

Inter-quartile Range:-

It measures the variability of a distribution by giving us the range covered by the middle 50% of the data.

The IQR is difference between quartile (Q_3) and quartile Q_1 .

Steps to calculate:-

Step 1:-

1. Arrange the data in ascending order.
2. Find the median M .
3. Recall that the median divides the data.
4. So, that 50% of data points are below

the ~~mean~~ median.

5. Remaining 50% of the data points are above the median.

$$2\sqrt{2} = 2$$

Step 2:-

1. Find the median of the lower 50% of data
2. This is called 1st quartile of distribution
3. The point is denoted by Q_1
4. Note that Q_1 divides the lower 50% of the data into two halves.
5. It contains 25% of data points in each half.
- 6.

Step 3:-

1. Repeat this again for the top 50% of the data.
2. Find the Median M and followed the

Step 1.

Step 4:-

then find the IQR, $IQR = Q_3 - Q_1$

Example:-

Find IQR

2	1	Bottom half	7	4
2	5 6 6 6 9		7	
3	6 1 3 3 3 3 4 4 4		8	0
3	5 5 5 7 8 9	TOP half		
4	1 1 1 2 3			
4	5 9 9			
5				
5				
6	1			
6				

$$Q_1 = (31 + 33) / 2 \Rightarrow 32$$

$$Q_3 = (41 + 42) / 2 \Rightarrow 41.5$$

$$IQR = 41.5 - 32 \Rightarrow 9.5$$

Problem

Find spearman's rank correlation coefficient for the following data:

Data x

Data y

7

50

3

19

20

80

9

55

11

66

14

72

1

4

4

36

12

70

3

35

Soln

Data x in ascending order : 1, 3, 3, 4, 7, 9, 11, 12, 14, 20

Rank X : 1, 2.5, 2.5, 4, 5, 6, 7, 8, 9, 10

Data y in ascending order : 4, 19, 35, 36, 50, 55, 66, 70, 72, 80

Rank y : 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Data x	Data y	Rank x	Rank y
1	50	5	5
3	19	2.5	2
20	80	10	10
9	55	6	6
11	66	7	7
14	72	9	9
1	4	1	4
4	36	4	8
12	70	8	3
3	35	3	

d	d ²
0	0
0.5	0.25
0	0
0	0
0	0
0	0
0	0
0	0
0	0
0	0
0.5	0.25

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \times 0.5}{10(10^2 - 1)}$$

$$= 1 - \frac{3}{99} = 1 - 0.00303$$

$$\Rightarrow 0.997$$

Least Square Regression Model :-

Monday

1. To find the best fit for N points

Step 1:- for each (x, y) point calculate x^2 and xy

Step 2:- Sum all x, y, x^2 and xy , which gives as

$\Sigma x, \Sigma y, \Sigma x^2$ and Σxy ($\Sigma \rightarrow$ means "sum up")

Step 3:- calculate slope m

$$m = \frac{N \Sigma (x, y) - \Sigma x \Sigma y}{N \Sigma (x^2) - (\Sigma x)^2}$$

$N \rightarrow$ No. of points

Step 4:- calculate Intercept

$$b = \frac{\Sigma y - m \Sigma x}{N}$$

Step 5:- Assemble the eqn of line.

Problem:-

x
Hours of Sunshine

Ice cream sold

2

4

3

5

5

7

7

10

9

15

Solr

It shows positive Linear regression.

Step 1:-

x	y	x^2	xy
2	4	4	8
3	5	9	15
5	7	25	35
7	10	49	70
9	15	81	135
$\Sigma x = 26$	$\Sigma y = 41$	$\Sigma x^2 = 168$	$\Sigma xy = 263$

Step 2:-

Step 3:-

$$m = \frac{5 \times 263 - 26 \times 41}{5 \times 168 - 26^2}$$

$$= \frac{1315 - 1066}{840 - 676} \Rightarrow \frac{249}{164} = 1.5183$$

Step 4:-

$$b = \frac{41 - 1.5183 \times 26}{5} = 0.3409$$

Step 5:-

$$y = mx + b$$

$$y = 1.518x + 0.3405$$

x	y	Predicted value	Error
2	4	3.34	-0.66
3	5	4.86	-0.14
5	7	7.9	+0.9
7	10	10.93	+0.93
9	15	13.97	-1.03